

Varun Taneja

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EDUCATION

University of Notre Dame | Notre Dame, IN

GPA: 3.81/4.0, Dean's List

Bachelor of Science (Major: Computer Engineering, Concentration: Cybersecurity).

Graduation: May 2025

WORK EXPERIENCE

University of Notre Dame CSE Department | Notre Dame, IN

January 2023 – May 2025

Logic Design Teaching Assistant

- Facilitated student comprehension of Boolean algebra, Assembly language, and Verilog (office hours and studio sessions.)

Research Assistant, Ye's Lab

July 2024 – August 2024

- Collaborated on "[Multi-objective Personalized Health-aware Food Recommendation System with LLM-enhanced Interpretation](#)" culminating in a paper submission to the KDD 2025 ADS Track.
- Developed and implemented baseline models (NCGF and HAFR) using Python and PyTorch benchmark lab's recommendation system.

Partior | Singapore

May 2024 – June 2024

Digital Intelligence Team Intern

- Generated a 12-month mock billing dataset with realistic growth; ingested into GCS for Looker Studio dashboards, enabling billing team to model seasonal revenue projections.
- Built a Python pipeline using Google's Gemini API & Vertex AI to extract business data from online PDFs, reducing manual entry effort with near-zero errors.
- Conducted comprehensive analysis of prompt strategies (zero-shot, prompt-chaining) to optimize data retrieval, helping sales team identify new lead segments.

Vena Energy | Singapore

May 2022 – June 2022

Data Analytics and Programming Intern

- Migrated SCADA data-processing code from Node.js to Python, consolidating data pipelines for ~10 wind and solar power plants and enabling more maintainable, scalable scripts.
- Developed and deployed a real-time Python application that pulled SCADA metrics from each plant and updated Google Firebase every minute—powering live health monitoring for all assets.
- Created a Python script to archive multi-year SCADA datasets into BigQuery on GCP, automatically handling tens of gigabytes of historical data to free up storage and ensure seamless dashboard queries.

Singapore Armed Forces | Singapore

October 2019 – July 2021

Guardroom In Charge and Service Role Security Trooper

- Led guardroom operations in partnership with Sergeants: managed SAR21 rifle issuance, camp clearances, scheduling, earning promotion to Corporal.
- Enforced military law by overseeing vehicle access, conducting systematic checks, and streamlining entry processes via ID-card camp passes.

LEADERSHIP AND SERVICE

Notre Dame Club Golf Team | Notre Dame, IN

August 2022 – May 2025

Vice President, Anchor Player (Handicap +2)

- Competed in regional NCCGA tournaments, earning four straight National Championship bids.
- Managed team practices, coordinated tournament travel squads, and oversaw all competitive aspects of the team.

Notre Dame Club Boxing (Bengal Bouts) | Notre Dame, IN

August 2022 – May 2025

Competitor & Fundraiser

- Competed in an amateur boxing tournament in February of 2023, 2024, and 2025.
- Raised over \$8000 to support education programs for students associated with Holy Cross Missions in Bangladesh.

RELATED COURSEWORK

NavigAIte: An AI-Enabled Travel Planner | University of Notre Dame

January 2025 – May 2025

- Conducted contextual inquiries with 5 users, mapped key pain points, and defined 3 AI-driven itinerary modes.
- Designed a JSON-schema/chain-of-thought LLM framework for on-demand, personalized travel recommendations
- Built full-stack solution (Next.js/React frontend, Python backend) integrating travel APIs & Llama-3.3-7b for real-time, budget-aware itineraries.

SKILLS AND INTERESTS

Programming Languages: Advanced C/C++, Python, NextJS, Tailwind CSS, CSV, JSON, Verilog, Git, MIPS, RISC-V, Scikit, Pytorch

Technical: Advanced Apple Suite, Microsoft Office, GCP, Firebase, Cadence Virtuoso, Ollydbg, IDA Pro

Language: English (fluent), Hindi (conversational). **Interests:** Golf, Cooking, Rock Music, Fitness, Boxing